

- Q.28 Describe the methods for improving the duty of irrigation water.
- Q.29 What do you mean by super critical and sub critical flow.
- Q.30 What are the criteria for selecting of irrigation pumps.
- Q.31 Explain adaptability and limitation of sprinkler system.
- Q.32 What do you understand by tube wells.
- Q.33 What are the methods of improving duty.
- Q.34 What are the criteria for selection of irrigation pumps.
- Q.35 What is aquifer constant.

SECTION-D

Note: Long answer questions. Attempt any two questions out of three Questions. (2x10=20)

- Q.36 Explain various water conveyance structures. Open channels, their types and layouts.
- Q.37 Describe the operation and maintenance of submersible pump, their common troubles and remedies.
- Q.38 What is evapotranspiration. Explain its need and importance for its measurement. List various methods for its estimation / measurement.

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3rd Sem.

Branch : Agri

Sub: Farm Irrigation Engineering/ Water Res. Engg.

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple type Questions. All Questions are compulsory. (10x1=10)

- Q.1 Total depth of irrigation a crop is known as
a) Duty b) Delta
c) Base d) Evapotranspiration
- Q.2 Centrifugal force is involved in a generation of pressure in :
a) Axial flow pump b) Radial flow pump
c) Mixed flow pump d) Reciprocating pump
- Q.3 Permeability is expressed in.
a) Meter per second
b) Meter
c) Meter² per second
d) Volume percentage of pore space
- Q.4 Leaching efficiency per unit of water is highest for
a) Boarder irrigation b) Basin irrigation
c) Furrow irrigation d) Sprinkler irrigation

- Q.5 The duty is largest
- At the head of water course
 - On the field
 - At the head of main canal
 - Same at all places
- Q.6 The properties of semi confined aquifer may be expressed in terms of.
- Storage coefficient
 - Transmissibility
 - All of the above
 - Hydraulic resistance
- Q.7 Centrifugal force is involved in the generation of pressure in
- Axial flow pump
 - Radial flow pump
 - Reciprocating pump
 - Mixed flow pump
- Q.8 Well development is a process of
- Lower casing pipe
 - Drilling of tube well
 - Reversals of flow through casing pipe
 - Gravel packing
- Q.9 Sprinkler irrigation is recommended in
- Highly undulating area
 - Soils having high infiltration rate
 - Irrigation source in a tube well
 - In all the above situation
- Q.10 Coefficient of storage is a property of
- Confined aquifer
 - Unconfined aquifer
 - Perched aquifer
 - None of the above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What are the merits of irrigation.
- Q.12 Explain any two types of impellers.
- Q.13 List two benefits of canal lining.
- Q.14 List two methods to improve duty.
- Q.15 Types of open well (Names only)
- Q.16 List two limitations of drip irrigation system.
- Q.17 What is flood irrigation.
- Q.18 What is intensity of irrigation.
- Q.19 What is delta and duty.
- Q.20 What is consumptive use of water.

SECTION-C

Note: Short answer type Questions. Attempt any twelve questions out of fifteen Questions. (12x5=60)

- Q.21 Explain the necessity of irrigation in a tropical country like INDIA.
- Q.22 What are the various types of irrigation.
- Q.23 What do you understand by spillways.
- Q.24 Describe a brief note on sprinkler system.
- Q.25 Explain any two methods to improve duty.
- Q.26 Explain any two design elements of open wells.
- Q.27 What is siphon spillway.

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